Skills Development for Adolescent Girls

Investment case, East Java, Indonesia

Hernando Grueso, PhD · UNICEF

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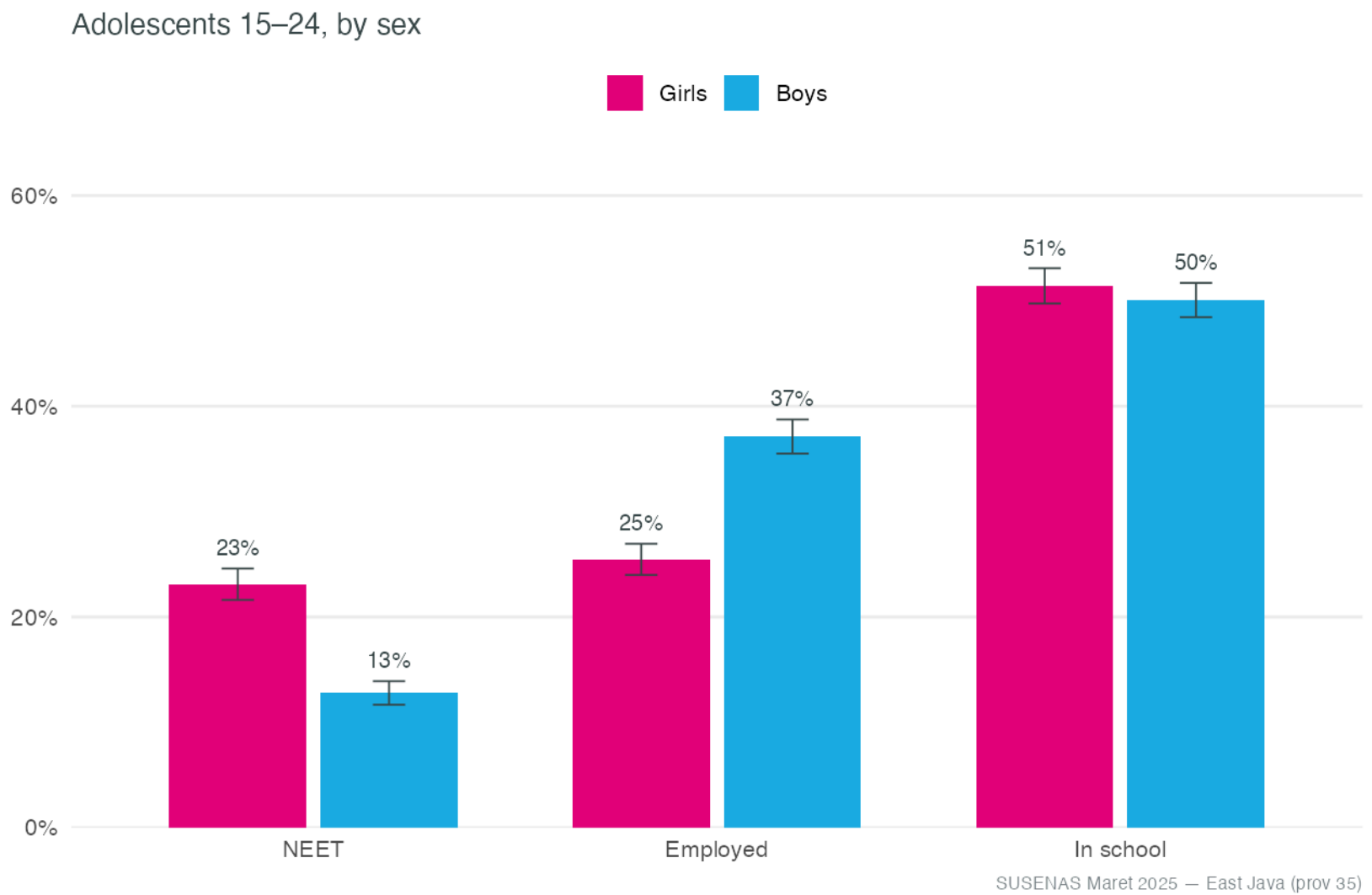
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Appendix: regression results, methods and data

1 · The problem

Girls have the same school attendance rate as boys, but their NEET rate is close to double

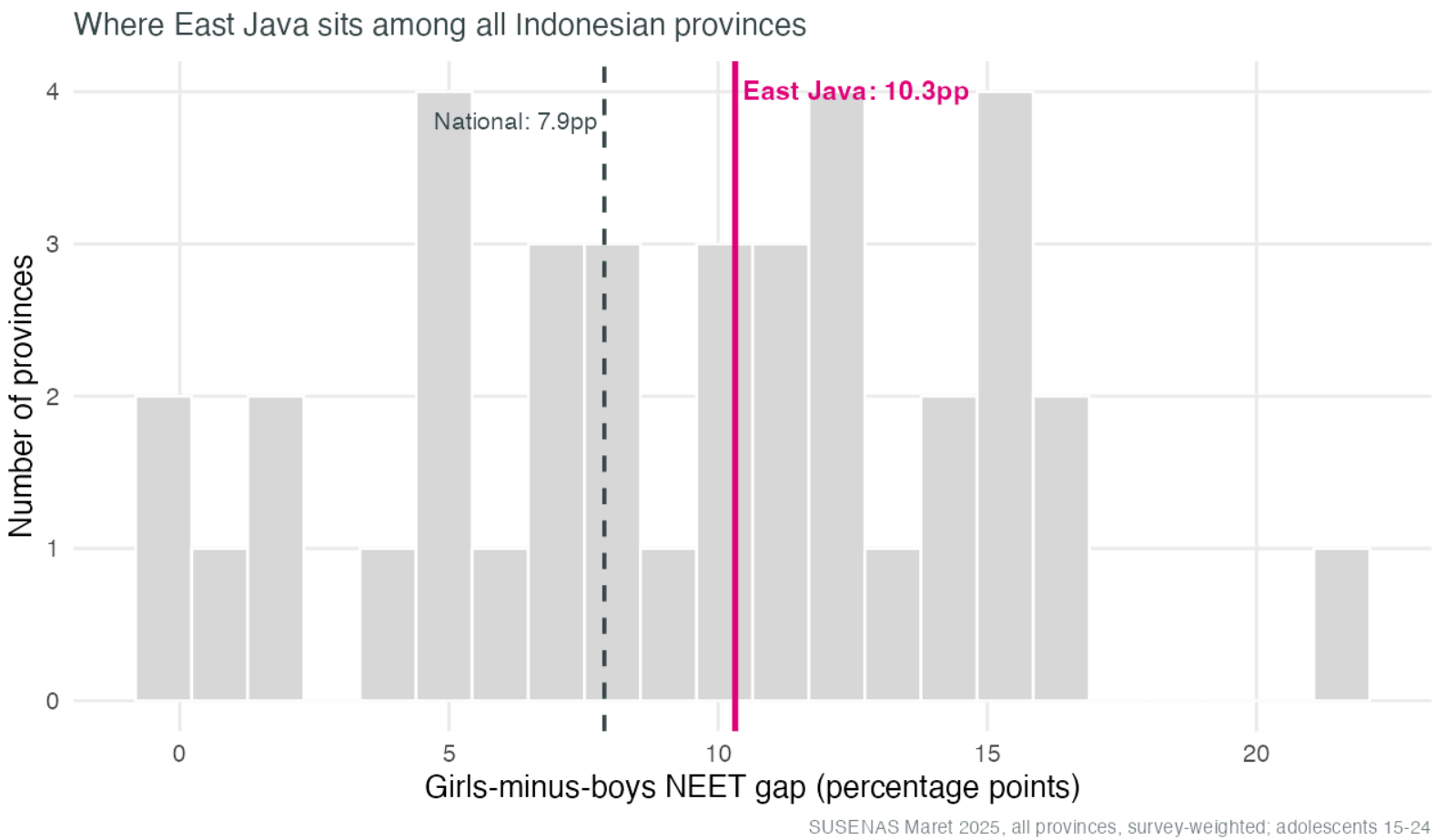


Observed: SUSENAS 2025

NEET: not in employment, education or training. Girls 15–24: 51% in school (boys 50%), 23% NEET (boys 13%), 25% employed (boys 37%). The gap opens at the transition out of school. SUSENAS Maret 2025, survey-weighted, 95% CIs.

East Java’s NEET gender gap is wider than the national average

Observed: SUSENAS 2025, all provinces



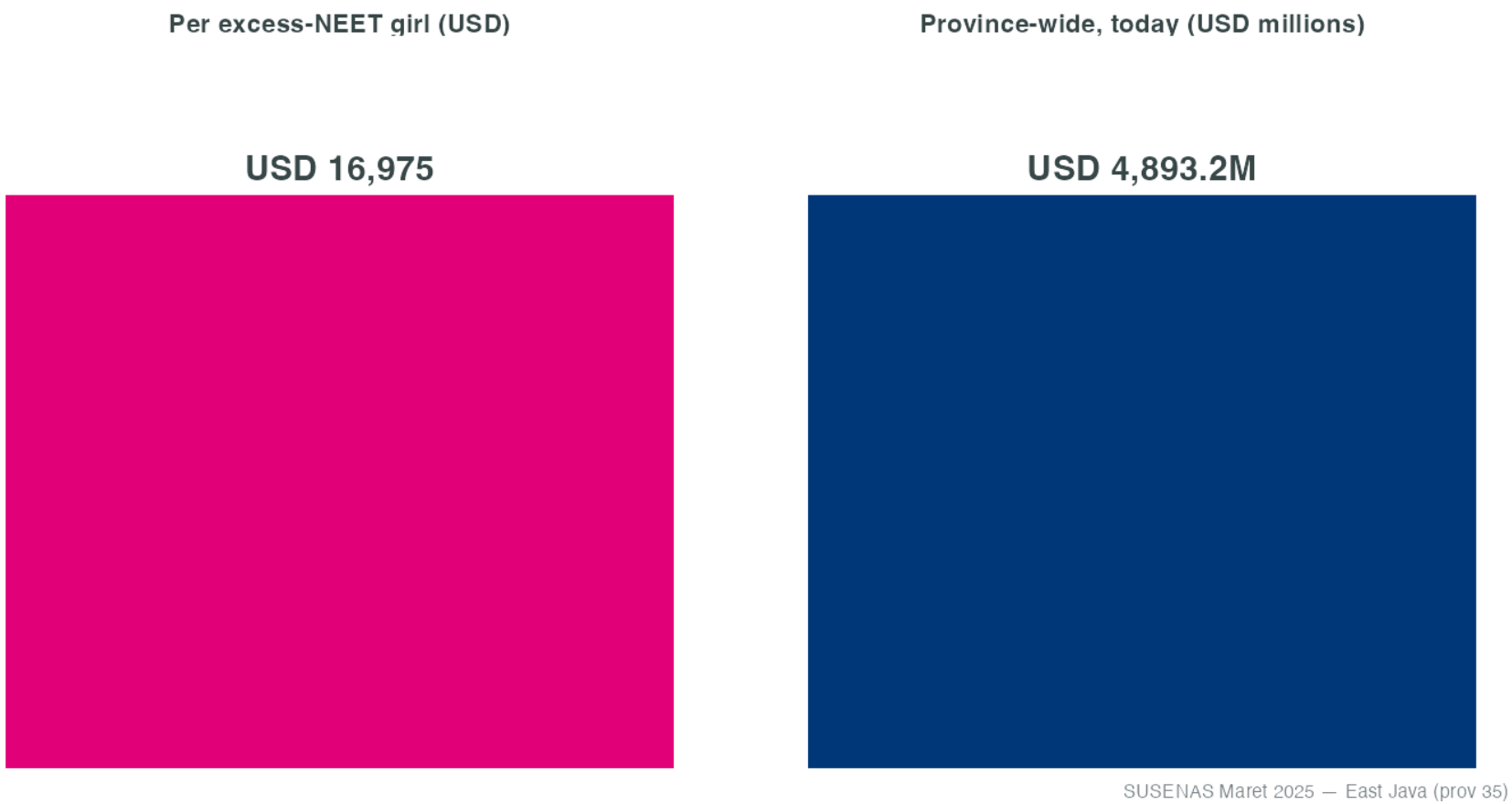
Girls-minus-boys NEET gap: 10.3 points in East Java against 7.9 points nationally. Same survey, same definition, all provinces.

The gap costs an estimated USD 4.9 billion in forgone lifetime productivity

Estimate: SUSENAS 2025 with earnings assumptions

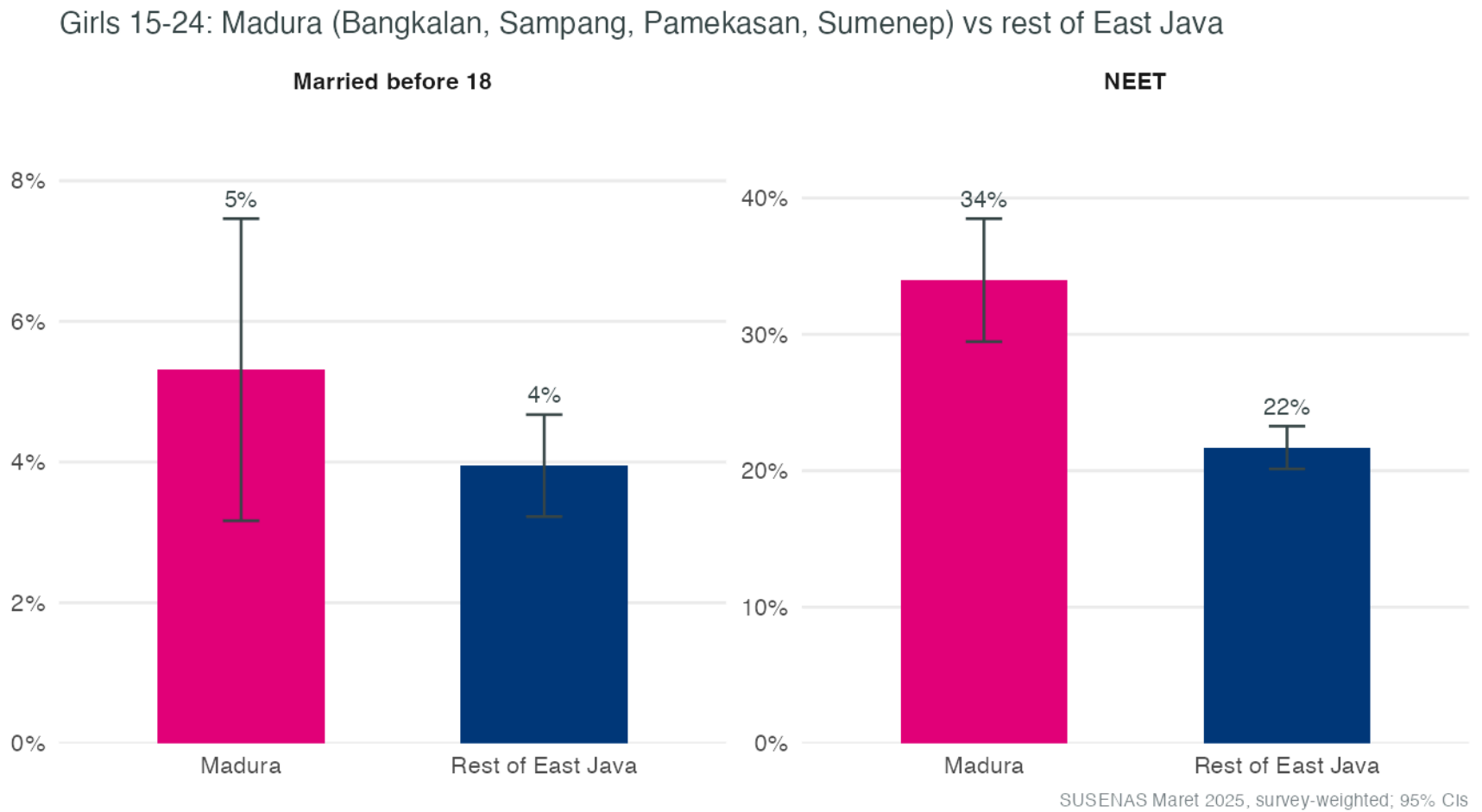
Forgone lifetime productivity from the NEET gender gap

Excess NEET girls (10.3pp gap above the boys' rate) x present value of lifetime earnings



About 288,000 girls are NEET in excess of the boys' rate. Each would be expected to earn about USD 17,000 more over a working lifetime (present value) if she had the boys' NEET rate.

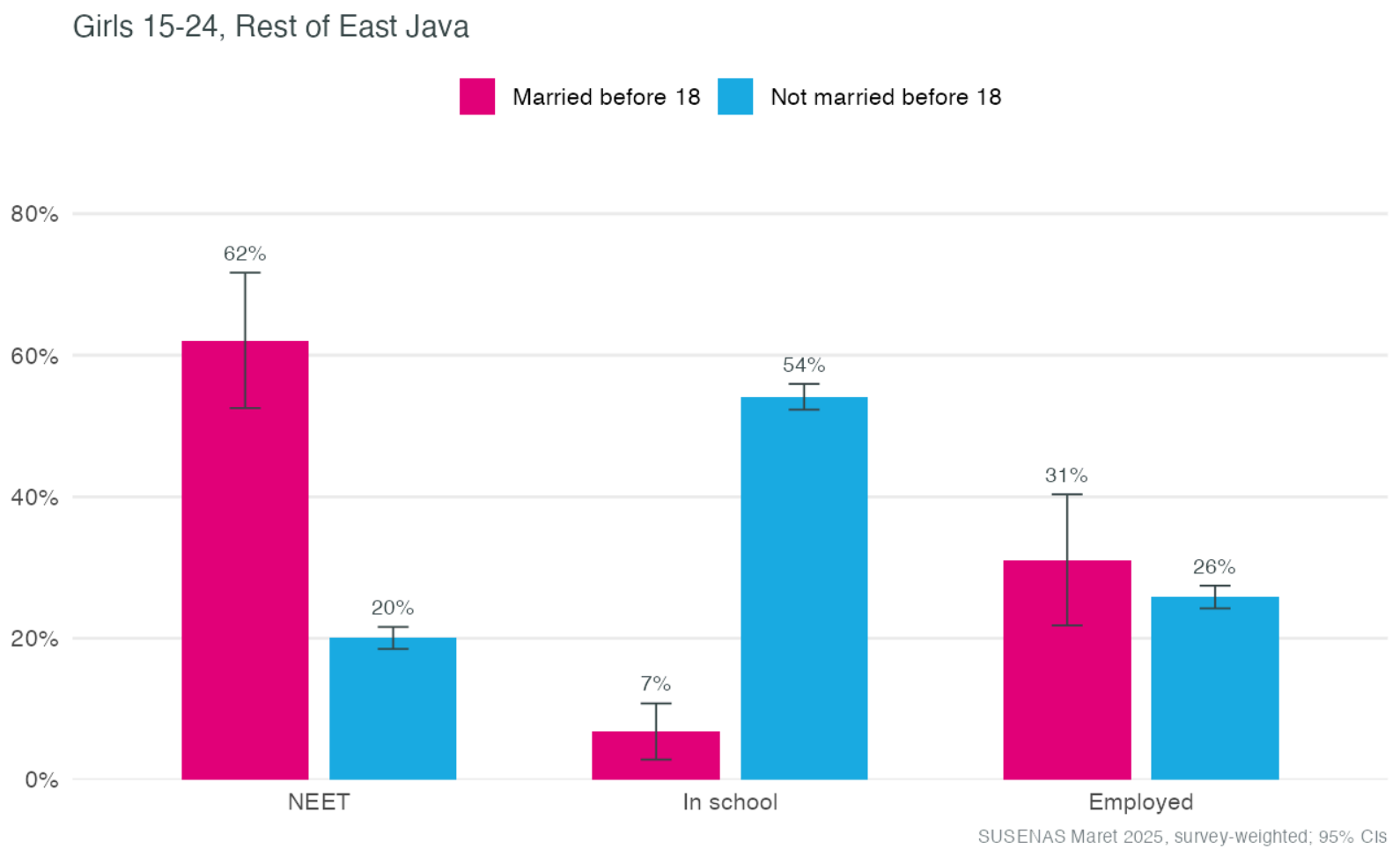
Madura has higher NEET for girls and boys, with a similar gender gap



Observed: SUSENAS 2025

Girls' NEET is 34% in Madura (Bangkalan, Sampang, Pamekasan, Sumenep) against 22% in the rest of East Java; boys' NEET is 23% against 12%. The gender gap is similar (11 vs 10 points), and fewer girls are in school (45% vs 52%). Madura is a priority area for a pilot.

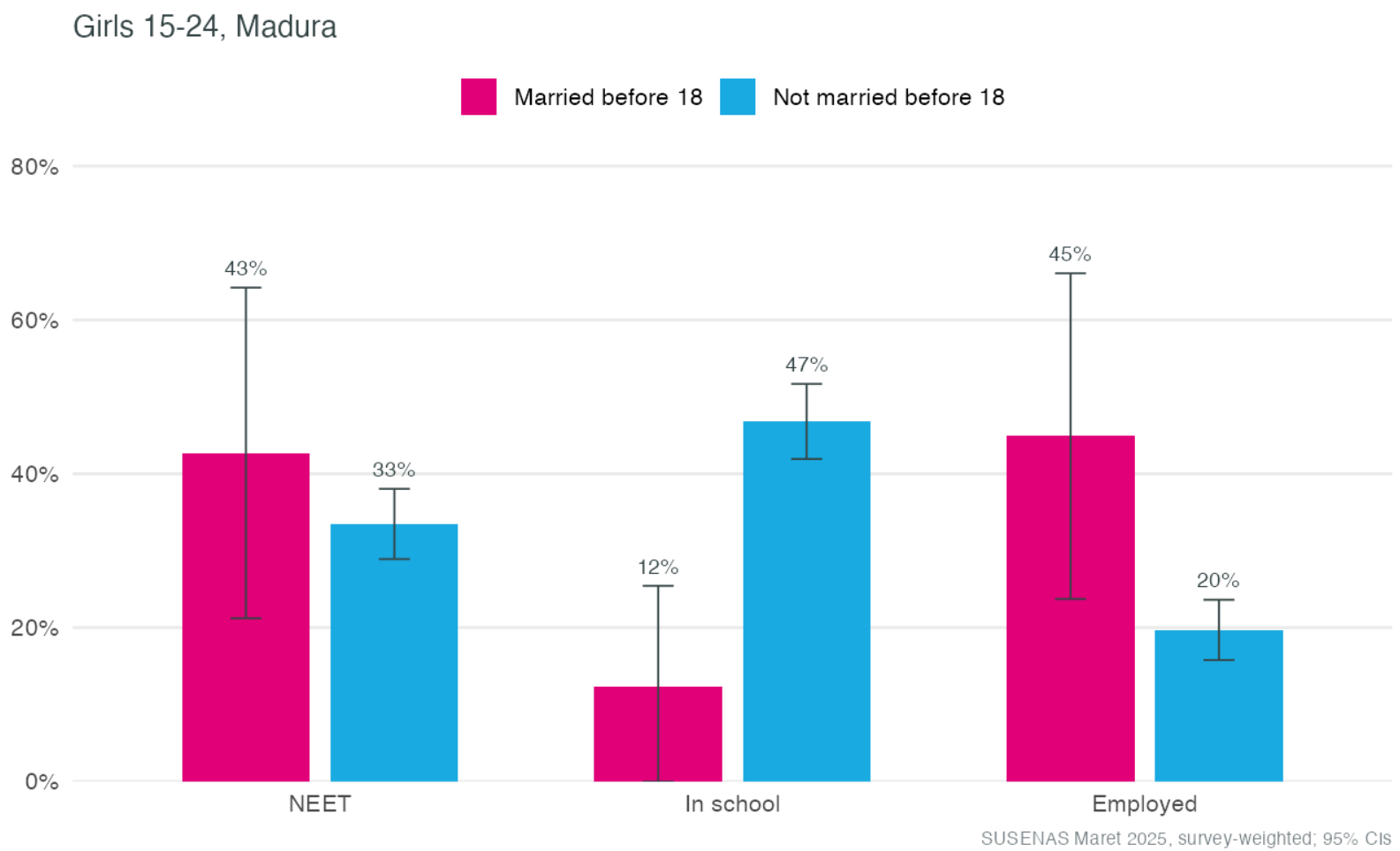
Child marriage and outcomes: rest of East Java



Observed: SUSENAS 2025

Girls 15–24 outside Madura, by whether they married before 18, survey-weighted (95% CIs). Regression tests by region are in the appendix. Early pregnancy before 18 overlaps almost entirely with child marriage in this data, so only child marriage is shown. Association, not causation.

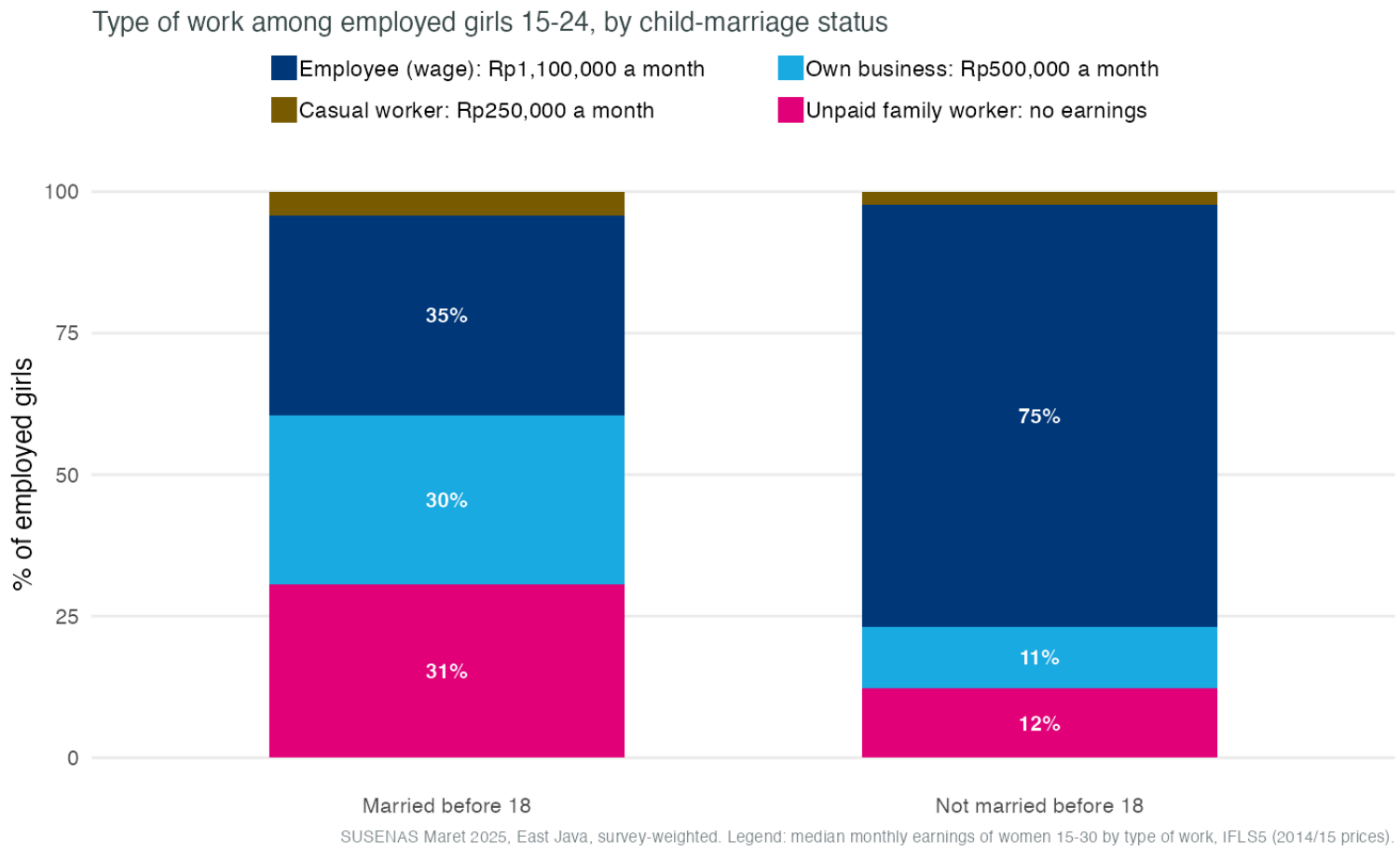
Child marriage and outcomes: Madura



Observed: SUSENAS 2025

Girls 15–24 in Bangkalan, Sampang, Pamekasan and Sumenep. Fewer girls, so wider intervals. Association, not causation.

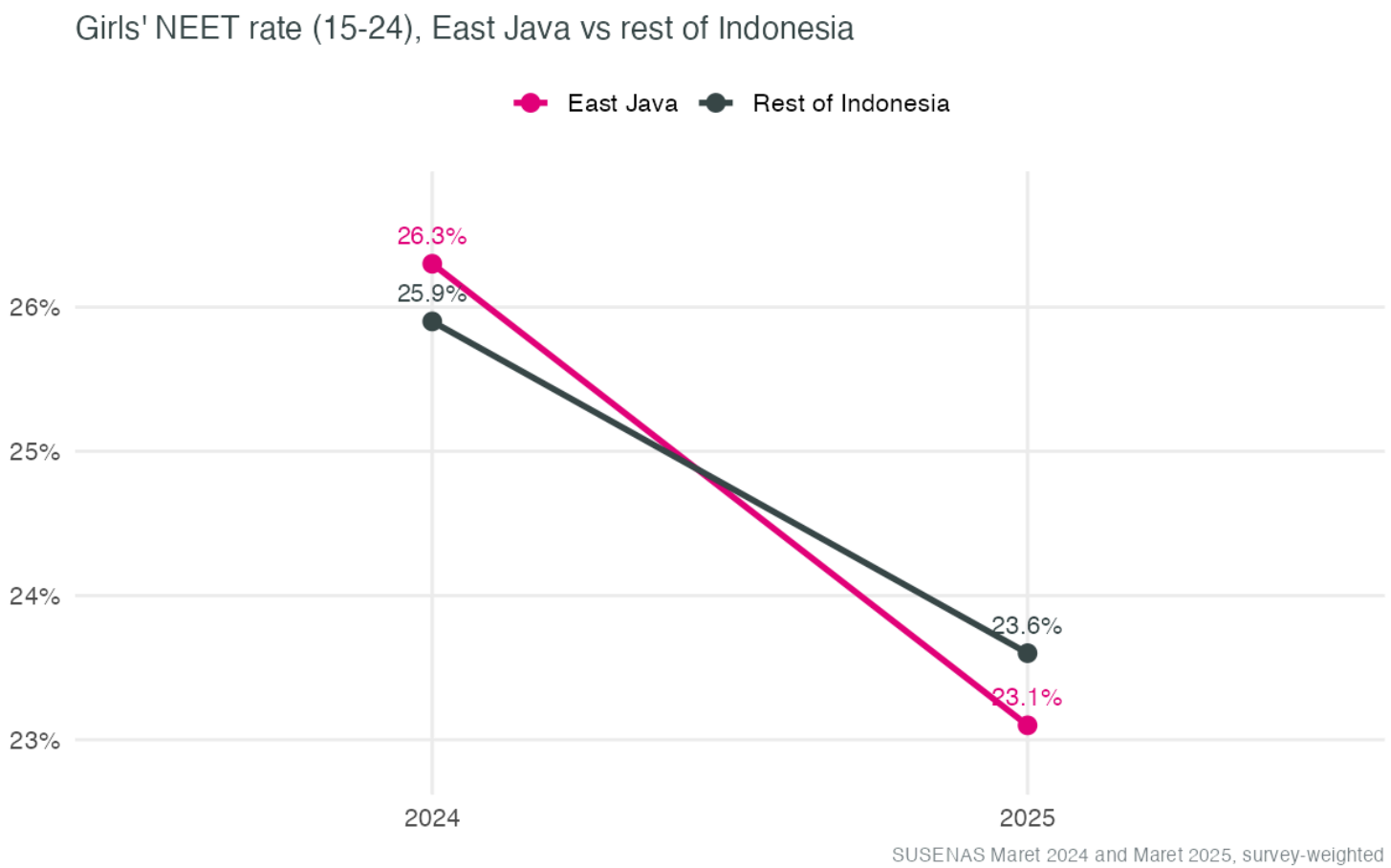
Married girls who work do more unpaid family work and less wage employment



Observed: SUSENAS 2025 and IFLS5

Among employed girls married before 18, 35% are wage employees (others 75%) and 31% are unpaid family workers (others 12%); 22% work under 20 hours a week (others 12%). Based on 78 employed married girls, so indicative. SUSENAS has no earnings; the appendix shows earnings by type of work for young women from IFLS5.

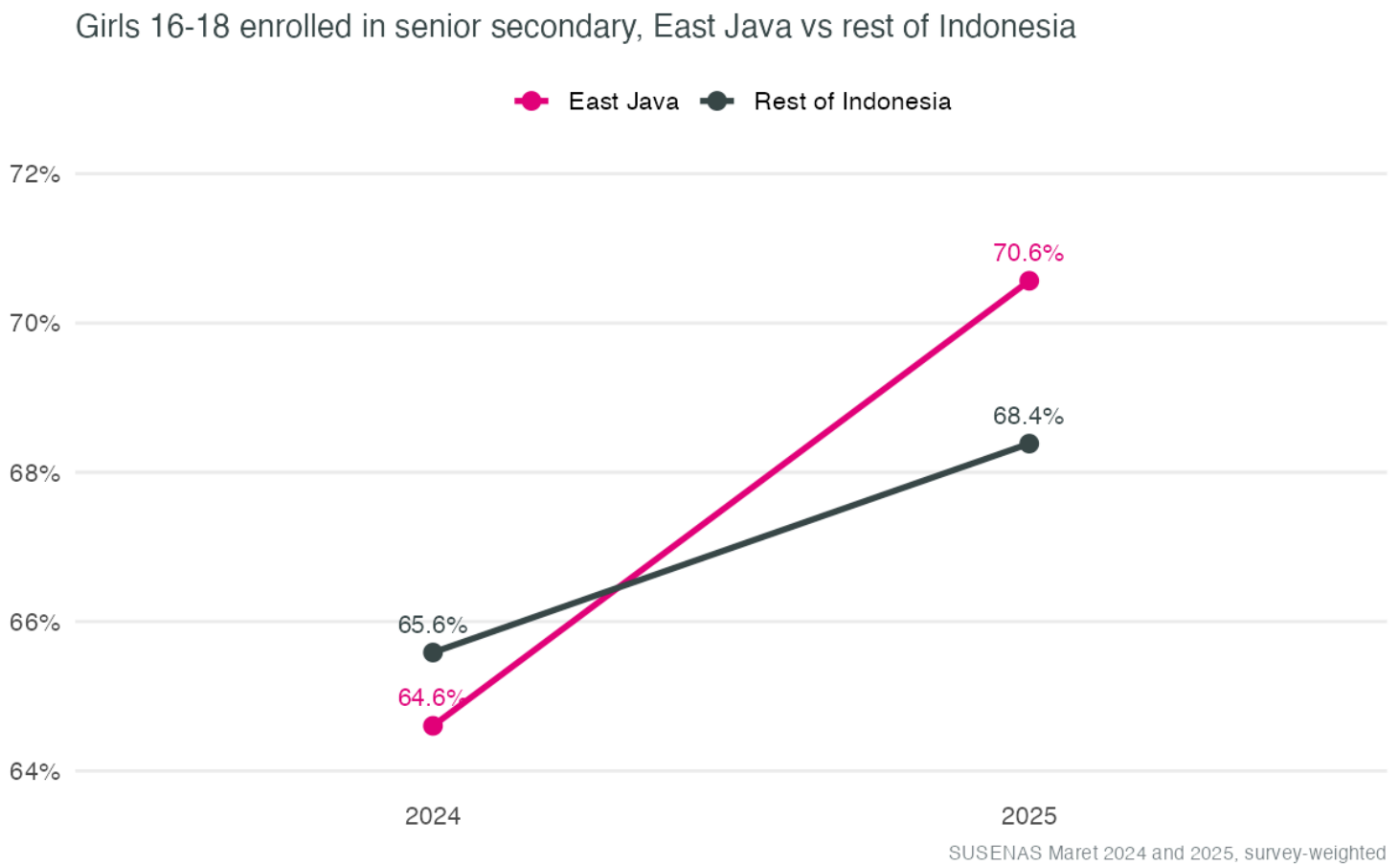
Girls’ NEET in East Java and the rest of Indonesia, 2024 and 2025



Observed: SUSENAS 2024 and 2025

Girls’ NEET (15–24) fell 3.2 points in East Java and fell 2.3 points in the rest of Indonesia; child marriage fell 1.3 points and fell 0.7 points; tertiary enrolment at 18–20 rose 3.2 points and rose 5.0 points. Both years use the same definition (any work in the last week, or temporarily absent from a job; not attending school). Survey trends cannot separate a programme from wider changes; a comparison of programme and non-programme districts is pending.

Senior-secondary enrolment rose faster in East Java than in the rest of Indonesia



Observed: SUSENAS 2024 and 2025

Share of 16–18 year olds enrolled in senior secondary rose 6.0 points for girls in East Java against 2.8 in the rest of Indonesia (boys: 5.8 against 2.6). The faster rise applies to both sexes, so it points to something affecting East Java's schools generally rather than a girls-specific programme. The district comparison can test whether programme districts drove it.

What changed for girls between 2024 and 2025

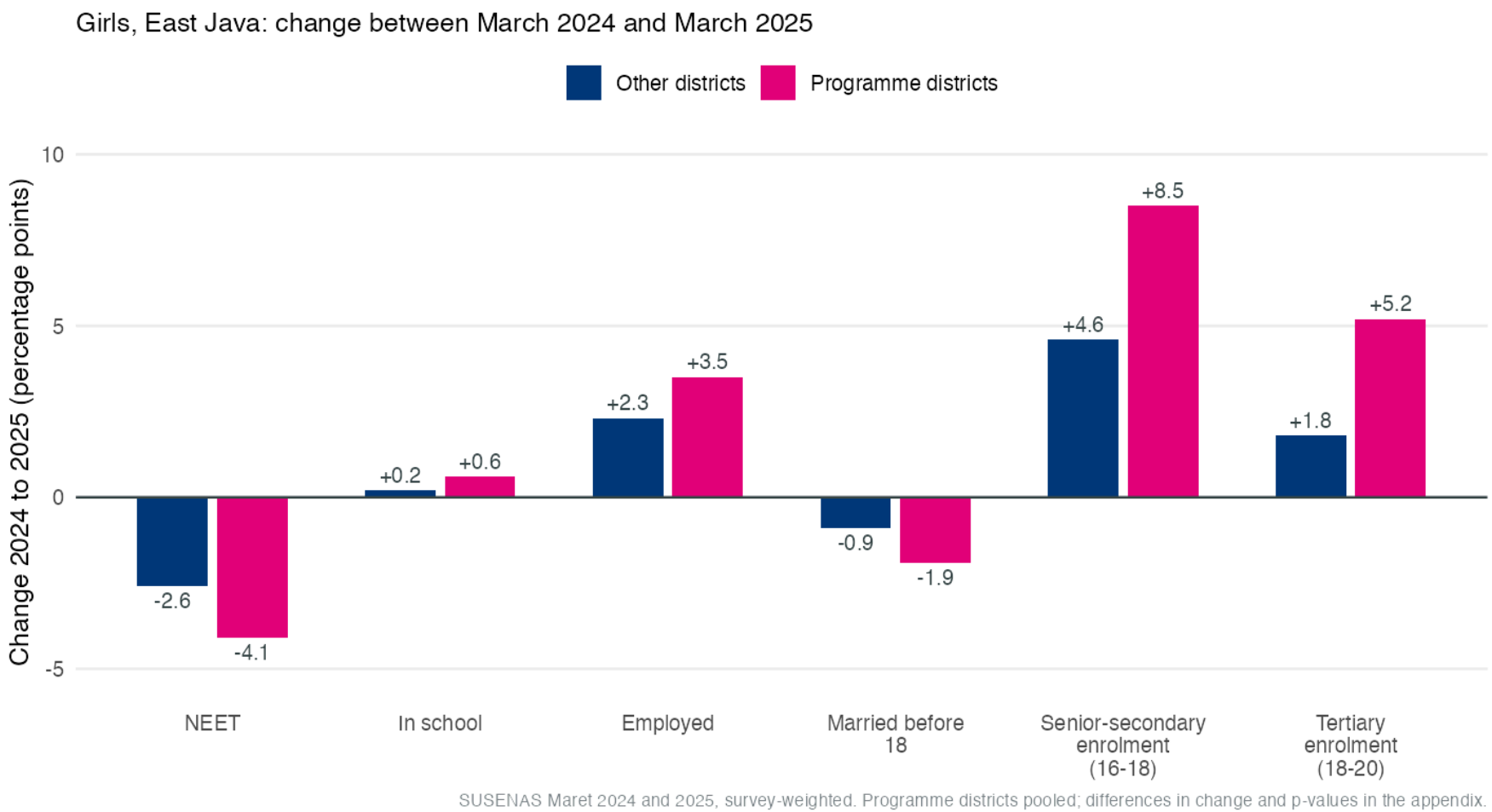
Observed: SUSENAS 2024 and 2025

Indicator	East Java 2024	East Java 2025	East Java change	Rest of Indonesia change
NEET, 15-17	9.5	6.8	-2.8	-1.3
NEET, 18-21	28.6	24.7	-3.9	-3
NEET, 22-24	40.7	37.5	-3.2	-2.2
NEET, 22-24 with tertiary certificate	32.7	34.3	1.6	-1.1
NEET, 22-24 without tertiary certificate	42.7	38.5	-4.3	-2.5
In school, 15-24	51.1	51.4	0.4	0.5
Employed (not in school), 15-24	22.7	25.5	2.8	1.7
Own business, % of employed	12.6	11.8	-0.8	-0.2
Employee, % of employed	71.6	72.5	0.9	-0.1
Unpaid family worker, % of employed	13.3	13.3	0	0.6
Trade or transport sector, % of employed	27.9	32.4	4.5	-0.5

Girls 15–24, same definitions in both years. Reading the table against the explanations raised: the national internship programme would show up as a fall concentrated among 22–24 year olds with a tertiary certificate; growth in informal and gig work as a rising share in own business or in trade and transport; schooling policies as a rising share in school.

Programme districts compared with the rest of East Java

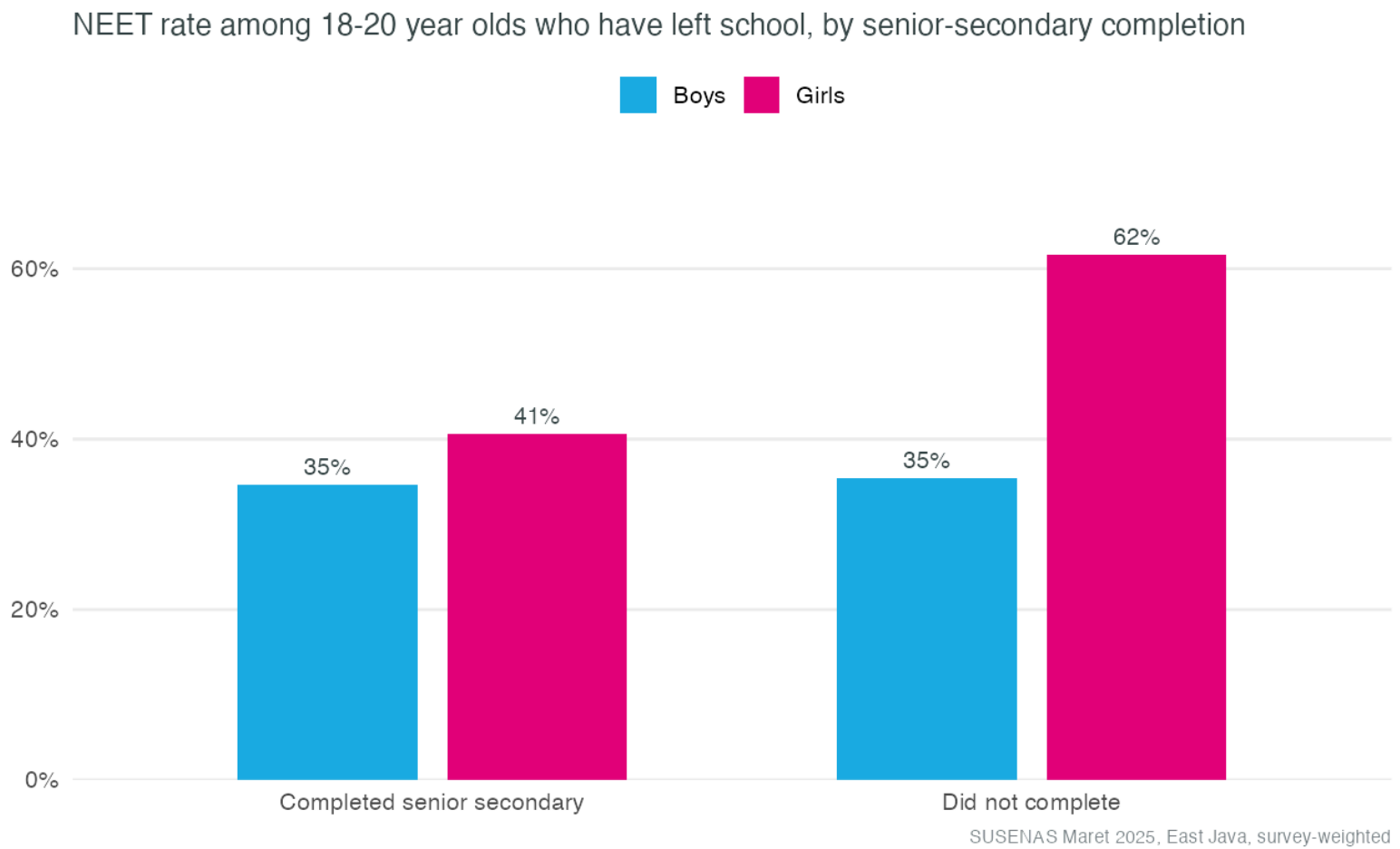
Observed: SUSENAS 2024 and 2025



Twelve programme districts pooled, against the other 26. Difference in the change in girls' NEET: -1.5 points (p = 0.487). Programme districts were selected rather than assigned at random, and if the programme started before March 2024 part of its effect is already in the baseline, so this is a descriptive comparison, not an impact estimate.

2 · Theory of change

Completing senior secondary is associated with 20 points lower NEET for girls



Observed: SUSENAS 2025

Among 18–20 year olds who have left school, girls who completed senior secondary have a NEET rate of 41%, against 62% for those who did not. Adjusted for age and area (girls 18–24), completion is associated with a 20-point lower NEET rate ($p<0.001$). For boys there is no observed difference. Keeping girls in school to completion is the programme's main lever.

Evidence from Indonesia

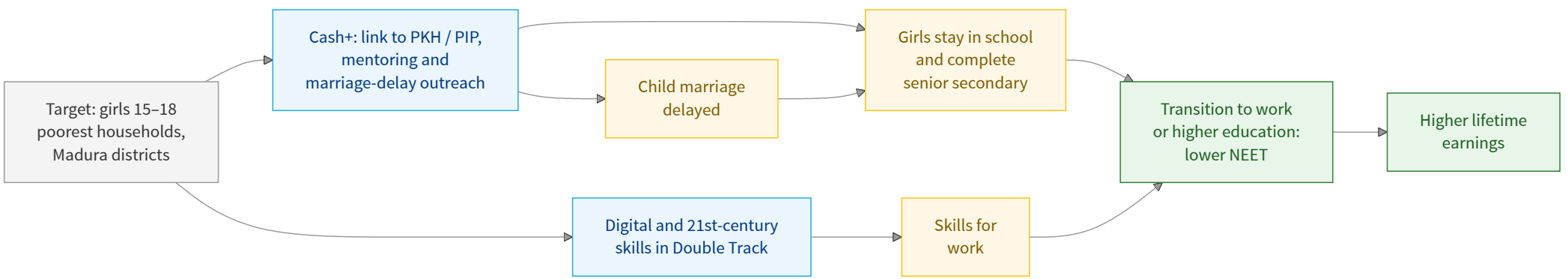
Published evidence

Programme	Design	Result	Relevance
PKH cash transfer (Priebe & Sumarto, 2025)	Quasi-experimental	Child marriage −3.5pp among girls	A cash component reduces child marriage
PIP education assistance (Ulfa & Rezki, 2024)	Matching and IV	School participation +11.4pp (pooled, not girls-specific)	Anchors the effect used in the ROI, with a transferability assumption
PIP, girls' senior secondary (Caniago et al., 2021)	Observational regression	Girls' senior-secondary enrolment +24.6pp (adjusted association)	Targets the stage where girls drop out
Vocational vs general secondary schooling (Pritadrajati, 2022)	Quasi-experimental	Female LFPR +3.4pp (significant at 10%); income +5.6% (not significant)	Used only as an uncertain scenario in the ROI

In SUSENAS 2025, girls in PKH households have lower child-marriage rates than other girls (3% vs 4–6%), even though PKH targets poorer households. Consistent with the PKH evidence, though not a causal estimate.

Theory of change

Framework based on published evidence



3 · Projected returns (ex-ante)

Key assumptions

Modelled projection (ex-ante)

Assumption	Value	Source
Reduction in girls' NEET	7.4 points	PIP school-participation effect (+11.4pp, Ulfa & Rezki 2024) × 65% of girls who leave school before completing senior secondary being NEET (SUSENAS 2025)
Extra schooling for a girl kept in school	3 years	Senior-secondary cycle
Earnings gain per year of schooling	8.0%	IFLS5 wage return for women, controlling for region, parents' education and selection into paid work
Female earnings	Rp20 million/year	Below East Java minimum wage 2025, reflecting informal work
Probability of earning (employment rate, women 25–54, East Java)	62%	SUSENAS Maret 2025
Earnings growth / discount rate	2% / 3% per year	Career growth; 3% real, with 2% and 5% in sensitivity
Skills earnings premium (scenario only, not in the central case)	5.6% for 10 years	Point estimate for vocational vs general schooling (Pritadrajati 2022), not statistically significant
Programme cost	USD 200 per girl	USD 58 per year for 3 years, plus 15% overhead; Cash+ links to existing PKH/PIP

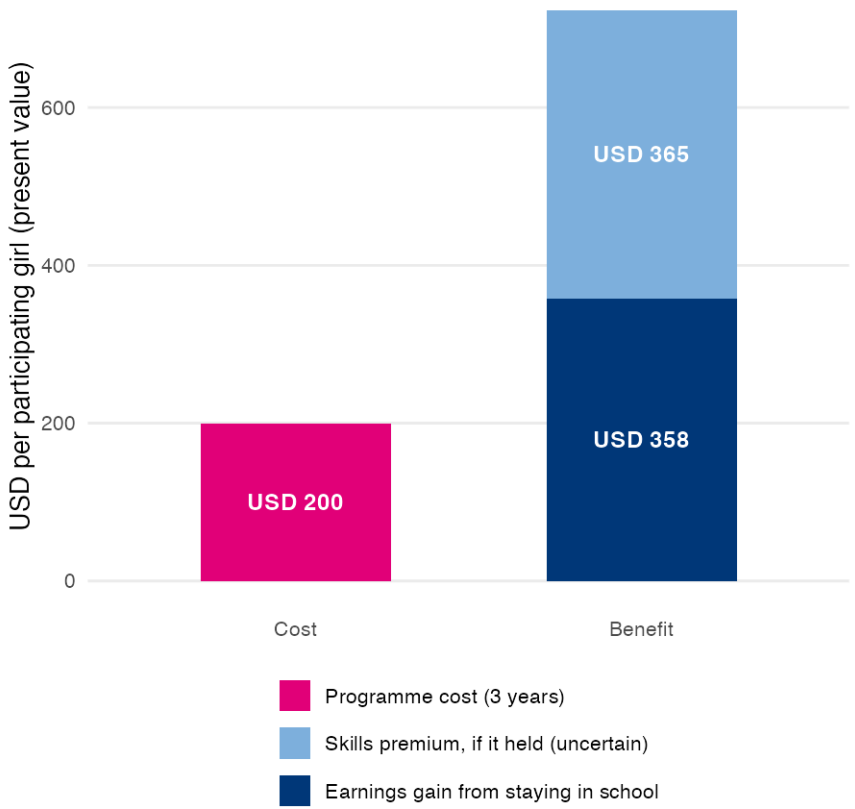
How the return is built

Modelled projection (ex-ante)

Step	Value
Extra years of schooling for a girl kept in school	3
Earnings gain per year of schooling	8.0%
Earnings gain for that girl	24%
Female earnings (USD/yr) x labour force participation	1,266 x 62%
Extra earnings per year (USD)	188
Present value over the working life (USD per girl moved out of NEET)	4,843
Girls exposed per girl moved out of NEET (1 / NEET reduction)	14
Cost per girl moved out of NEET (USD)	2,710
Return per USD 1, retention channel (central)	1.79
Skills premium of 5.6% for all exposed girls who work, if it held (USD per exposed girl, PV)	365
Return per USD 1 if the skills premium held	3.61

Every USD 1 invested could return up to USD 3.61 in lifetime earnings

Modelled projection (ex-ante)



- **Both channels together: USD 3.61 per USD 1.** USD 723 in lifetime earnings per girl against a cost of USD 200.
- **Conservative view, staying in school only: USD 1.79 per USD 1.** USD 358 per girl, based on Indonesian causal evidence (PIP). The programme breaks even if it reduces girls' NEET by 4.1 points; the evidence suggests 7.4. At the 2025 minimum wage this rises to USD 2.48.
- **Skills channel on its own: USD 1.82 per USD 1.** USD 365 per girl if the 5.6% earnings premium for vocational schooling in Indonesia applies; that estimate is not statistically significant, so this is the less conservative part.
- **Not counted in either:** delayed marriage, health, children's outcomes and fiscal returns. These are benefit-cost ratios, not fiscal returns to government.
- **The USD 200 per girl** covers three years of digital content, teacher training, mentoring, shared connectivity and marriage-delay outreach, plus 15% overhead; PKH and PIP transfers are existing entitlements.

Benefit-cost ratio by scenario

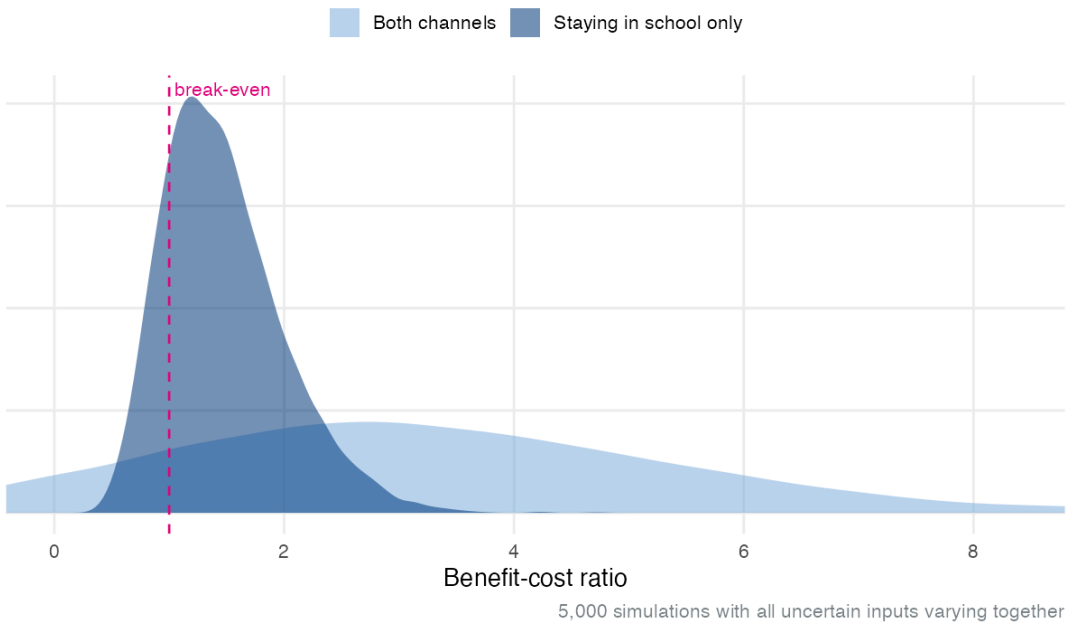
Modelled projection (ex-ante)

Scenario	Retention benefit (USD/girl)	Skills premium benefit (USD/girl)	Total benefit (USD/girl)	Cost (USD/girl)	BCR
Central: retention channel (Indonesian causal evidence)	358	0	358	200	1.79
Plus skills earnings premium of 5.6% (point estimate, not statistically significant)	358	365	723	200	3.61
Retention channel, earnings at the 2025 minimum wage	496	0	496	200	2.48
Retention channel, with Cash+ stipend for the poorest 40%	358	0	358	359	1.00

The central case is the retention channel; the second row adds the uncertain skills premium; the others show earnings at the 2025 minimum wage, and the cost of adding a Cash+ stipend for the poorest 40%. With other assumptions varied one at a time, the central return stays between USD 1.22 and USD 2.74 (appendix).

How likely is the programme to pay for itself?

Modelled projection (ex-ante)



- **Staying in school only:** pays for itself in 82% of simulations; median return USD 1.40 per USD 1 (90% of results between 0.76 and 2.44).
- **Both channels:** pays for itself in 82% of simulations; median USD 2.98 (between -0.50 and 7.09).
- **What varies:** the PIP effect (±50%); the pass-through, return to schooling and skills premium with their estimated standard errors; earnings, employment, discount rate and cost over their ranges.

4 · Key messages

Key messages

1. Girls in East Java have the same school attendance rate as boys but are close to twice as likely to be NEET, a wider gap than nationally, costing about USD 4.9 billion in forgone lifetime productivity.

2. Completing senior secondary is associated with 20 points lower NEET for girls; child marriage with leaving school and more unpaid family work. NEET is highest in Madura, for girls and boys.

3. A USD 200-per-girl package returns at least USD 1.79 per USD 1 from keeping girls in school, and up to USD 3.61 if the skills component raises earnings, before counting delayed marriage or health benefits. Varying all assumptions together, it pays for itself in 82% of simulations on schooling alone.

Policy recommendations

1. Target the transition, not enrolment. Girls attend school at the same rate as boys, and senior-secondary enrolment rose faster in East Java than elsewhere in 2024–2025, but tertiary enrolment rose more slowly. Set targets on completion and the first job or further study.

2. Start in Madura and the poorest households. NEET is highest and attendance lowest in Madura, and child marriage falls with household wealth (2024 data).

3. Link girls to PKH and PIP rather than creating a new transfer. Indonesian evidence shows PKH reduces girls' child marriage; linkage uses existing systems.

4. Fund pedagogy and mentoring, not devices. Internet use is already near universal, and devices alone did not raise learning in Uruguay.

5. Measure the programme's own effect. Survey trends cannot separate the programme from wider changes. Compare programme and non-programme districts, and track whether girls move into wage employment rather than unpaid family work.

Appendix

Programme districts: comparison of changes

Outcome	Programme 2024	Programme 2025	Other 2024	Other 2025	Difference in change (pp)	SE	p
NEET	26.9	22.8	25.9	23.3	-1.5	2.1	0.487
In school	49.8	50.4	51.9	52.1	0.3	2.4	0.901
Employed	23.3	26.8	22.3	24.6	1.2	2.1	0.569
Married before 18	6.2	4.3	4.9	4	-1	1	0.331
Senior-secondary enrolment (16-18)	59	67.5	67.9	72.5	4	4.1	0.334
Tertiary enrolment (18-20)	21.5	26.7	21.2	23	3.3	3.9	0.396

Girls, East Java, survey-weighted (%). Difference in change: programme districts' change minus other districts' change, from a pooled survey-weighted regression with PSU-clustered standard errors.

Regression results: child marriage and outcomes

Hypothesis	Effect (pp)	SE	p	N
Child marriage → NEET	28.7	4.6	0.0000	6060
Child marriage → in school	-27.5	3.0	0.0000	6060
Child marriage → employment	-1.2	4.6	0.7942	6060

Linear-probability models, girls 15–24, survey-weighted, PSU-clustered. **Controls: age and urban/rural.** Poverty is not controlled for because the 2025 expenditure module is not yet available. The association with employment is small and not statistically significant. Association, not causation.

Child marriage and outcomes by region: regression results

Region	Outcome	Effect (pp)	SE	p	N
Madura	Employed	18.7	11.1	0.093	664
Madura	NEET	3.1	12.3	0.801	664
Madura	In school	-21.8	9.8	0.026	664
Rest of East Java	Employed	-4.6	5.1	0.361	5396
Rest of East Java	NEET	33.2	5.0	0.000	5396
Rest of East Java	In school	-28.5	3.1	0.000	5396
All East Java	Employed	-1.2	4.6	0.794	6060
All East Java	NEET	28.7	4.6	0.000	6060
All East Java	In school	-27.5	3.0	0.000	6060

Linear-probability models, girls 15–24, survey-weighted, PSU-clustered; controls for age and urban/rural. Effects in percentage points.

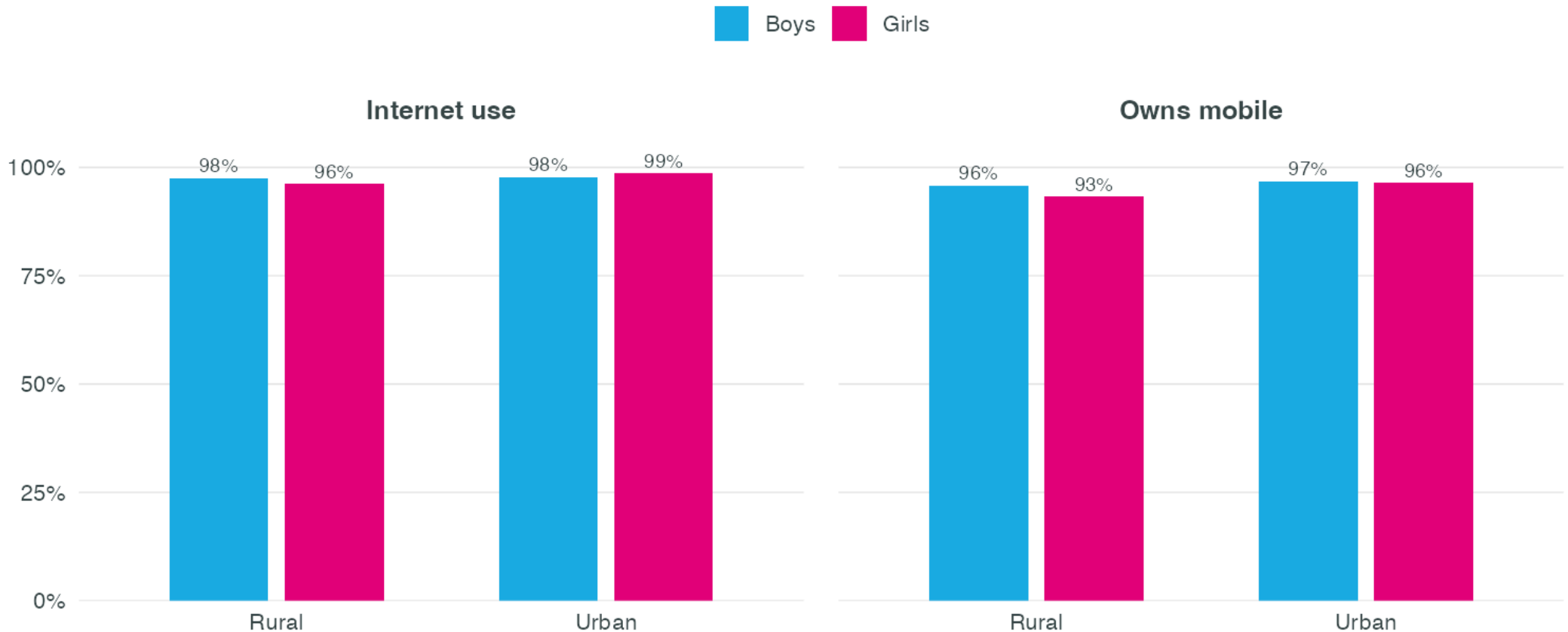
Earnings by type of work, young women (IFLS5)

Type	Women 15-30	With earnings reported	Median monthly earnings (IDR)
Employee (wage)	1662	1652	1,100,000
Own business	689	667	500,000
Casual worker	203	203	250,000
Unpaid family worker	614	614	0

Women 15–30, IFLS5 (2014/15 prices). SUSENAS collects no earnings. Wages for employees and casual workers; net profit for own businesses; unpaid family workers have no earnings by definition.

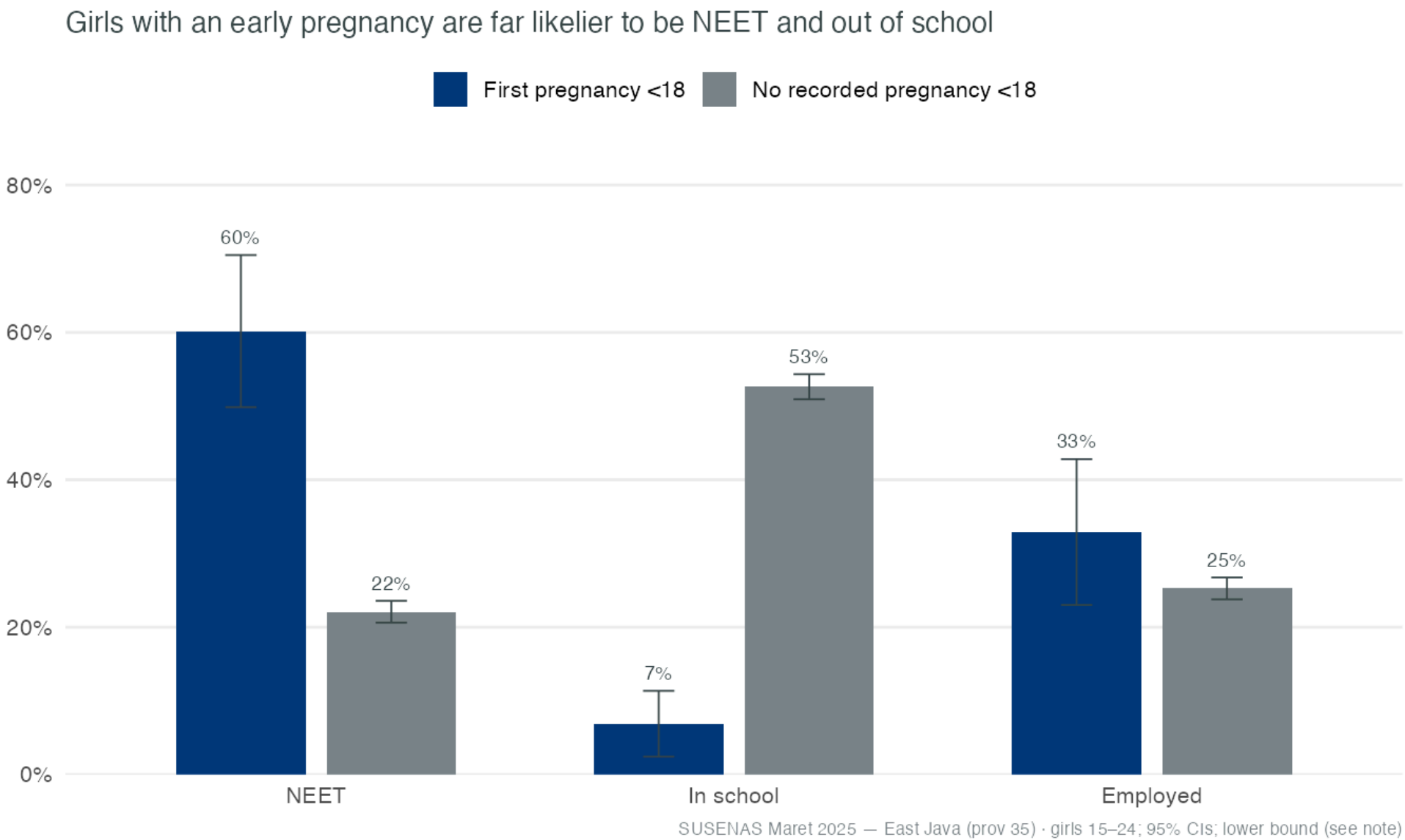
Digital access and use

Digital access and use, by sex and area



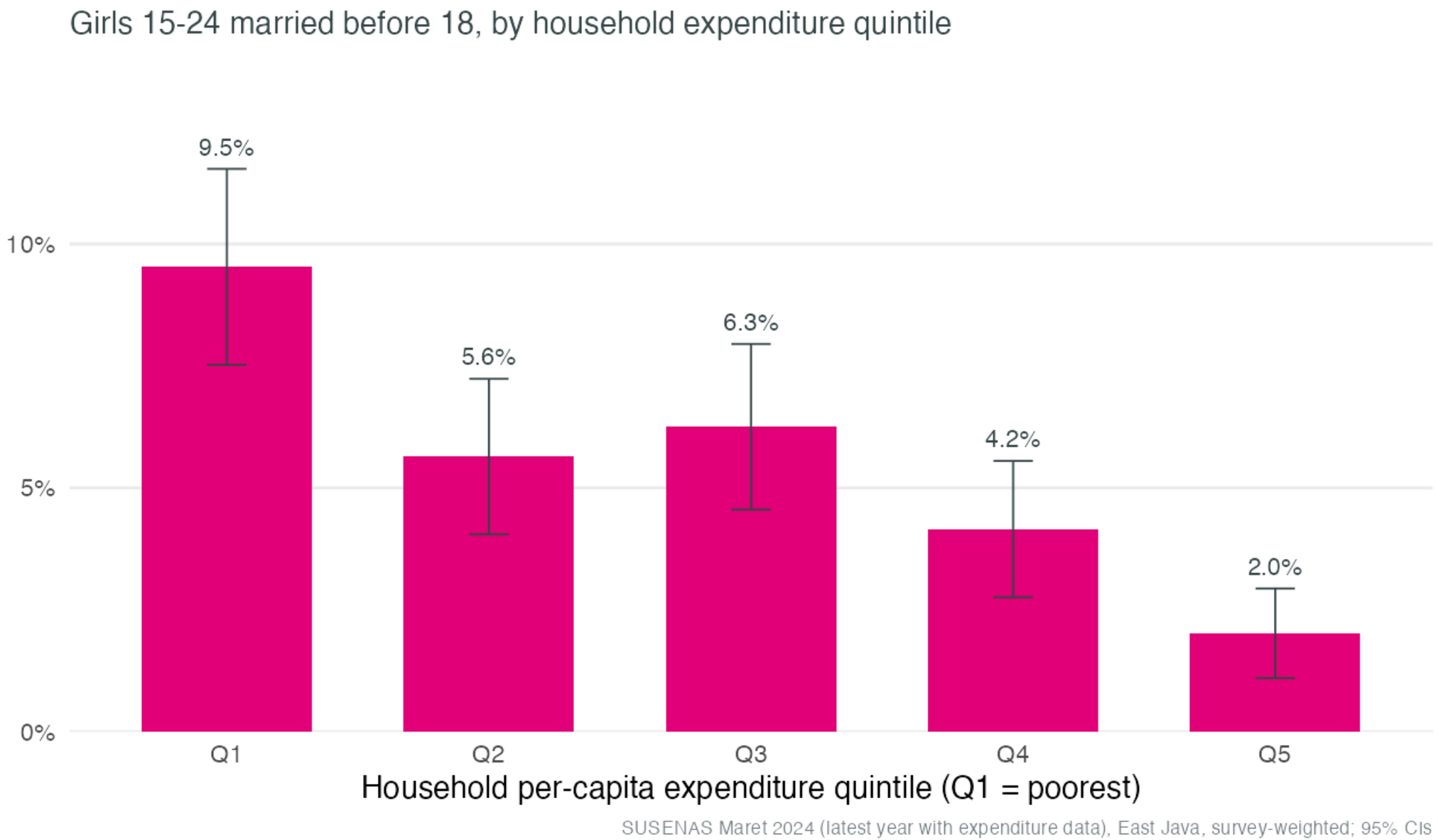
Source: SUSENAS Maret 2025 — East Java (prov 35). Weighted, adolescents 15–24.

Early pregnancy and outcomes



Recorded only for ever-married women, so a lower bound. Girls with a first pregnancy before 18 are almost all girls married before 18, so the main deck reports child marriage only.

Child marriage by household expenditure, 2024



SUSENAS 2024 is the latest year with household expenditure; the 2025 expenditure module has not yet been obtained.

Child marriage and NEET by household PKH receipt

region	PKH	n	Married_u18	NEET
Madura	Not PKH	509	6.1	35.9
Madura	PKH household	155	3.0	28.5
Rest of East Java	Not PKH	4661	4.1	22.1
Rest of East Java	PKH household	735	3.2	19.5

Girls 15–24, SUSENAS 2025. PKH targets poorer households, so lower rates among recipients are consistent with a protective effect, but are not a causal estimate.

Cost components

Component	USD/girl/yr	Source
Digital content & platform	10	Kemendikbudristek Merdeka Mengajar platform; INOVASI digital-learning unit costs
Teacher training (annualised)	15	Double Track PD days, annualised
Mentoring & socio-emotional	10	ELA/Skills4Girls facilitator costs
School connectivity/devices (shared)	15	Shared labs, no 1:1 devices
Ancillary: marriage-delay + flexible learning	8	ToR guidance: outreach to delay marriage, flexible options for caregivers
Annual direct cost per girl	58	Sum
Total per girl (3 years + 15% overhead)	200	Programme cost per girl

These are benchmark unit costs from comparable programmes, not quotes from Double Track; they are placeholders until the provincial government shares actual budgets. The transfer itself is not costed because the Cash+ component links girls to existing PKH and PIP entitlements; a new stipend for the poorest 40% at the PIP senior-secondary rate would add about USD 46 per girl per year (ROI table).

Sensitivity of the return to each assumption

Parameter	BCR at low value	BCR at high value
Skills earnings premium (all exposed girls who work)	1.79	3.61
NEET reduction per exposed girl	0.89	2.68
Mincer return per year of edu	1.26	2.74
Discount rate	1.22	2.20
Annual cost per girl (sensitivity)	2.30	1.38
Female annual wage (East Java)	1.34	2.23
Real earnings growth	1.33	2.10
Female labour force participation	1.29	1.86
Years the skills premium lasts	1.79	1.79

Each assumption varied on its own, others held at central values.

- **Low values**: a 2.5-point NEET effect (half the PIP-based estimate), the consumption-based return of 5.6% instead of 8%, earnings of Rp15m instead of Rp20m, a 5% discount rate, or a cost of USD 75 a year instead of 58.
- **High values**: a 7.5-point NEET effect, the 12.2% return without the selection correction, earnings of Rp25m, a 2% discount rate, or a cost of USD 45 a year.
- The effect size and the return to schooling matter most; no single assumption on its own takes the central case far from break-even.

What the return counts

Counted

Higher schooling leads to a lifetime earnings premium for girls moved out of NEET, using the female return to schooling over a discounted working life.

Excluded (so the return is a lower bound)

Delayed child marriage · health and productivity spillovers · intergenerational gains · fiscal returns · broader empowerment

Comparable international evidence

Study	Design	Sample	Finding	Relevance for East Java
Bergstrom & Özler 2021	Review of 104 interventions from 70 studies	Adolescent girls	Mixed evidence for life-skills training and girls' clubs; stronger for cash and schooling support	Supports cash and schooling components; cautions on clubs alone
Bandiera et al. 2018 (Sierra Leone)	RCT	Girls 12–25	Enrolment +8.5pp during the Ebola period; effects on income-generating activity vary by age	Girls' clubs protected schooling in a crisis context
Aramburu, Goicoechea & Mobarak 2021 (Argentina, Colombia)	RCT	Women 18–50	Coding bootcamps raised coding skills	Digital-skills content
Näslund-Hadley et al. 2025 (Paraguay)	RCT	Second-grade pupils	+0.09 SD computational thinking	Pedagogy; limited transfer to adolescent employment
Gülen et al. 2025 (Turkey)	Nonrandomized controlled pre/post	Girls 14–16	Higher STEM aspirations and motivation	Girls' STEM engagement
Prakash et al. 2019, Samata (India)	RCT	Poor girls 13–16	No overall effect on completion or marriage; schooling gains in one of two districts	Transition support alone had limited effects
Edmonds et al. 2021, Room to Read (India)	RCT	Girls	Dropout 4.1pp lower (main survey estimate)	Mentoring and life skills reduce dropout
Ashraf et al. 2025, Ten Years of Relational Power (Zambia)	RCT, 10-year follow-up	Girls ~14	+0.23 years of schooling	Durable, low-cost soft-skills layer
de Melo et al. 2014, Plan Ceibal (Uruguay)	Quasi-experimental	Pupils	No detectable maths or reading gains in the first two years	Devices alone did not raise learning; fund pedagogy
Faizah, Isbah & Azca 2025 (online; 2026 issue)	Qualitative	Indonesian youth	Norms, care duties and marriage drive girls' NEET	Indonesia-specific NEET context

Findings as reported in each source, with design and sample as verified. Used to inform programme design; the ROI effect comes from the Indonesian evidence. References on the last slide.

Returns to schooling: specification and results

Wage-based (IFLS5 2014/15, adults 20–60, estimated separately for women and men, weighted):

$$\ln(w_i) = \beta_0 + \beta_1 S_i + \beta_2 \mathrm{Exp}_i + \beta_3 \mathrm{Exp}_i^2 + \beta_4 \mathrm{Urban}_i + \gamma_p + \phi_f + \phi_m + \delta\lambda_i + \varepsilon_i$$

w: monthly wage; S: years of schooling; Exp: potential experience (age minus schooling minus 6); γ_p: province fixed effects; ϕ_f, ϕ_m: father's and mother's education (fixed effects); λ: selection term from a first-stage model of working, in which children under 5 in the household predict whether a woman works but not her wage (Heckman two-step).

Sex	Model	Return per year	SE	N
Women	(1) Baseline	12.0%	0.5	3830
Women	(2) + Region	12.2%	1.1	3830
Women	(3) + Family background	10.9%	1.1	3828
Women	(4) + Selection (Heckman)	8.0%	1.2	3828
Men	(1) Baseline	9.4%	0.3	6844
Men	(2) + Region	8.7%	0.6	6843
Men	(3) + Family background	8.0%	0.5	6842
Men	(4) + Selection (Heckman)	10.1%	1.0	6842

The ROI uses the women's estimate with all controls (model 4, 8.0%). The consumption-based return from SUSENAS 2024 (5.6% for women; household per-capita expenditure regressed on schooling, schooling^2, age, age^2 and urban) is the low case in the sensitivity analysis.

Data sources

- **SUSENAS Maret 2025** (BPS): NEET, school attendance and completion, child marriage, early pregnancy, type of work and hours, PKH receipt and digital access, for East Java and all provinces. East Java: 98,738 individuals, 12,599 aged 15–24, of whom 6,060 girls.
- **SUSENAS Maret 2024** (BPS): the 2024 comparison, and per-capita expenditure for the consumption-based return to schooling.
- **IFLS5 2014/15**: wages for the wage-based return to schooling.
- **Sakernas** (BPS): female labour force participation in East Java.
- **UMP Jawa Timur 2025**: benchmark for female earnings.
- **Published evaluations** of PKH, PIP and vocational training: the programme effect used in the return on investment.

All SUSENAS estimates are survey-weighted, with PSU/strata-clustered standard errors.

Definitions

- **NEET (15–24)**: not in employment, education or training, built as not attending school (R611; 2024: R610) and not working in the last week or temporarily absent from a job (R703_A; R704, 2024: R705).
- **Married before 18**: age at first marriage (R409) under 18.
- **Senior-secondary completion**: highest certificate (R615) at SMA/SMK/MA level or above.
- **Type of work**: status in main job (R706); hours (R707).
- **Madura**: Bangkalan, Sampang, Pamekasan, Sumenep.

Several variables were renumbered between the 2024 and 2025 questionnaires (school attendance R610 to R611, temporarily absent R705 to R704, certificate R614 to R615, employment status R707 to R706); the activity block (R703) is the same in both years and both years are coded identically. The operational NEET definition does not separately exclude participants in non-formal training, which SUSENAS does not record.

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